

BASICS OF ELECTRICAL ENGINEERING

(EE113ATD)



Unit-I

Introduction: Conventional and non-conventional energy resources; General structure of electrical power systems using single line diagram approach.

Power Generation: Hydel, Nuclear, Solar & wind power generation (Block Diagram approach).

DC Circuits: Ohm's Law and its limitations. KCL & KVL, applications to series, parallel and series-parallel circuits.

Introduction: Conventional and non-conventional energy resources

Sources of Energy

The two major sources of energy is classified as:

- Conventional Sources
- Non-Conventional Sources

The classification of the sources of energy is given in the below image.

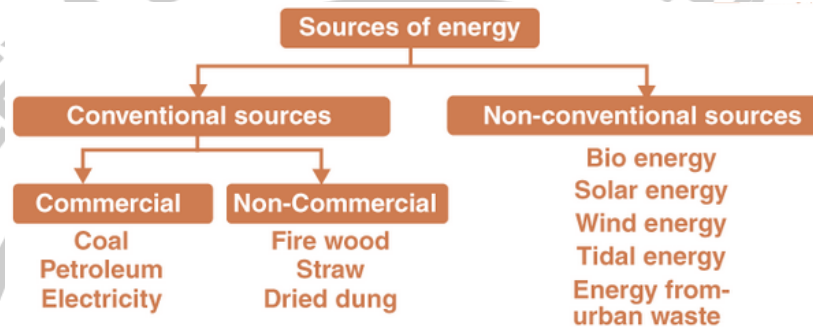


Figure 1-0.1 Classification of source of energy.

Conventional energy sources

Conventional energy sources, also known as non-renewable energy sources, are those that have been used for a long time and are available in limited quantities. They are typically formed over millions of years from the remains of ancient organisms. The most common examples include coal, oil, and natural gas. These sources are widely used for electricity generation, transportation, and heating.

Here's a more detailed look:

Fossil fuels:

Coal, oil (petroleum), and natural gas are the primary examples of conventional energy sources.

Nuclear energy:

While nuclear energy isn't a fossil fuel, it's also considered a conventional source because it's a long-established technology and uses non-renewable resources.

Hydropower:

Unit 1

While hydropower is often grouped with non-conventional sources due to its renewable nature (water is constantly replenished), some sources categorize it as conventional because it has been a long-standing energy source.

Non-renewable:

The key characteristic of conventional energy is that these resources are finite and cannot be replenished at a rate comparable to their consumption. Once used, they are gone.

Environmental impact:

Burning fossil fuels releases greenhouse gases, contributing to climate change and air pollution. Nuclear energy has its own environmental concerns related to waste disposal.

Commercial and non-commercial:

Conventional energy sources can be classified as commercial (like coal, oil, and natural gas used for electricity and transportation) and non-commercial (like firewood, which is often used in less developed regions).

These sources have been the backbone of global energy production for a long time, powering industries, transportation, and homes.

Non-conventional energy sources

Non-conventional energy sources, also known as renewable energy sources, are those that are continuously replenished by nature and can be used without depleting them. These sources are considered sustainable and environmentally friendly, offering an alternative to traditional fossil fuels. Examples include solar, wind, biomass, geothermal, and hydropower.

Here's a more detailed look at some key non-conventional energy sources:

Solar Energy: Harnessing the sun's radiant light and heat through solar panels to generate electricity or heat.

Wind Energy: Utilizing wind turbines to convert wind power into electricity.

Biomass: Generating energy from organic matter like wood, crops, and animal waste, either through burning or other processes.

Geothermal Energy: Tapping into the Earth's internal heat to produce electricity or heat buildings.

Hydropower: Generating electricity from the force of moving water, often through dams.

Ocean Energy: Utilizing the power of tides, waves, and ocean thermal gradients to produce energy.

These sources play a vital role in diversifying energy production, reducing reliance on fossil fuels, and mitigating climate change.

General structure of electrical power systems using single line diagram approach.

Electric power is the product of two quantities, current and voltage. These two quantities can vary with respect to time (AC power) or be kept at constant levels (DC power). Most refrigerators, air conditioners, pumps and industrial machinery use AC power, whereas most computers and digital equipment use DC power. The digital devices you plug into the mains typically have an internal or external power adapter to convert from AC to DC power). AC power has the advantage of being easy to transform between voltages and can be generated and utilized by brushless machinery. DC power remains the only practical choice in Digital systems can be more economical to transmit over long distances at very high voltages.

The ability to quickly transform the voltage of AC power is essential for two reasons: Firstly, power can be transmitted over long distances with less loss at higher voltages. So in power systems where generation is distant from the load, it is desirable to step up (increase) the voltage of power at the generation point and then step-down (decrease) the voltage near the load. Secondly, it is often more economical to install turbines that produce higher voltages than would be used by most appliances, so the ability to easily transform voltages means this mismatch between voltages can be easily managed Solid state devices, which are products of the semiconductor revolution, make it possible to transform DC power to different voltages, build brushless DC machines and convert between AC and DC power. Nevertheless, devices utilizing solid state technology are often more expensive than their traditional counterparts, so AC power remains in widespread use. Figure 1.2 shows Power Generation and Transmission and Distribution System.

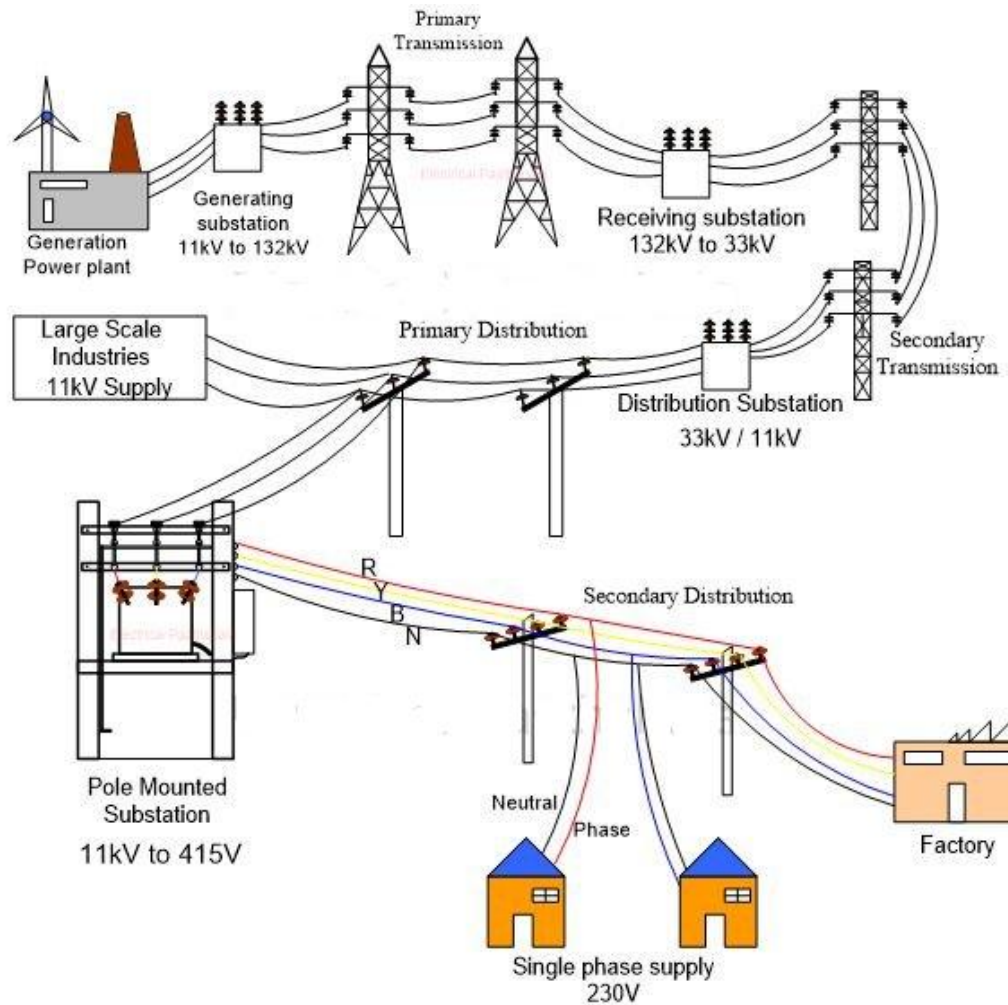


Figure 1-0.2 Generation, Transmission and Distribution of AC Electrical power.

Single-line diagram (SLD)

A single-line diagram (SLD) of India's power system visually represents the generation, transmission, and distribution networks using standardized symbols. It simplifies complex three-phase systems by representing a three-phase conductor with a single line, making it easier to understand the flow of power from generation to the end-user.

Components and Structure:

Generation:

The diagram starts with power generation sources like hydro, thermal, or renewable energy plants.

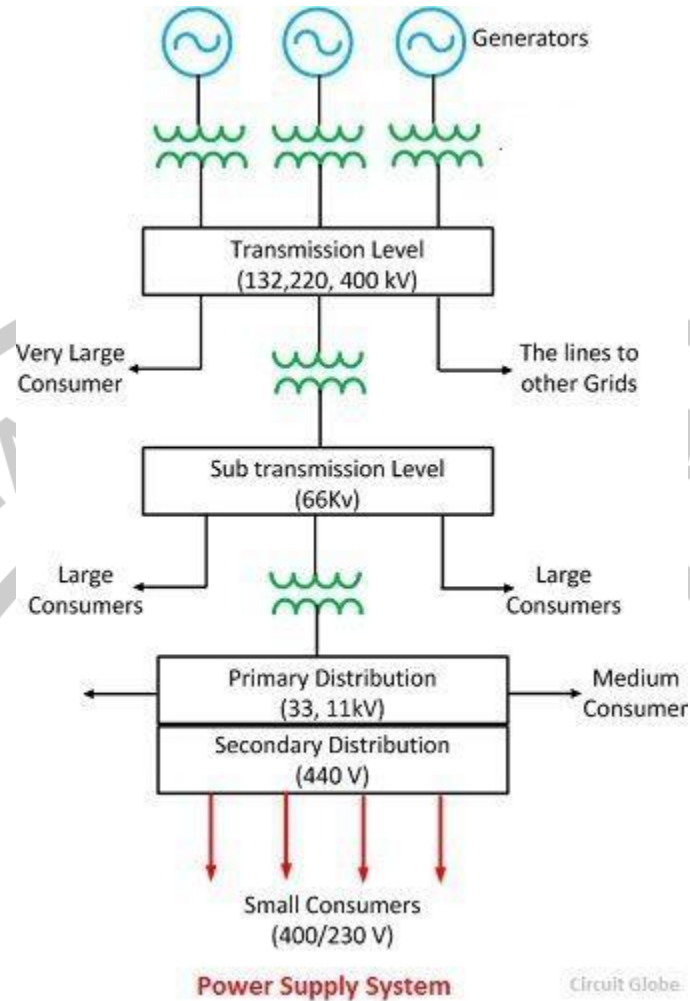


Figure 1-0.3 Single line diagram of electrical power system.

Transmission:

High-voltage transmission lines (e.g., 220kV, 400kV) carry electricity over long distances from generation stations to substations.

Substations:

Substations act as key nodes, stepping down the voltage for sub-transmission and distribution networks.

Sub-transmission:

Lower voltage lines (e.g., 66kV, 33kV) distribute power from main substations to local distribution substations.

Distribution:

Finally, low-voltage lines (e.g., 11kV, 400V) deliver power to residential, commercial, and industrial consumers.

Key Elements in the SLD:

Generators: Represented as alternators, converting mechanical energy into electrical energy.

Transformers: Used for stepping up or down voltage levels.

Circuit Breakers: Protect circuits from overloads and faults by interrupting the flow of current.

Busbars: Conducting bars that serve as common connection points for multiple circuits.

Transmission Lines: Represented by single lines, depicting the path of power transmission.

Loads: Represented by symbols indicating the end-users of electricity.

Significance of SLD:

Simplified Representation:

SLDs provide a clear and concise overview of a complex power system.

System Analysis:

They are essential for power flow studies, fault analysis, and other system studies.

Communication:

SLDs facilitate communication between engineers and other stakeholders involved in the power system.

System Planning and Design:

They aid in the planning and design of new power systems or modifications to existing ones.

Difference Between Conventional and Non-conventional Sources of Energy

Conventional Sources of Energy	Non-conventional sources of energy
These sources of energy are also known as a non-renewable source of energy	These sources of energy are also known as a renewable source of energy
They find both commercial and industrial purposes	They are mainly used for household purposes

These can be considered to be one of the reasons for the cause of pollution	These are not responsible for the cause of pollution
Coal, fossil fuels are two examples	Wind, solar energy and Biomass two examples

Power Generation

Hydel power generation

A hydroelectric power plant is a generating station which converts the potential energy of water at high level into electrical energy.

The schematic diagram of Hydro electric power plant is shown in the Fig.

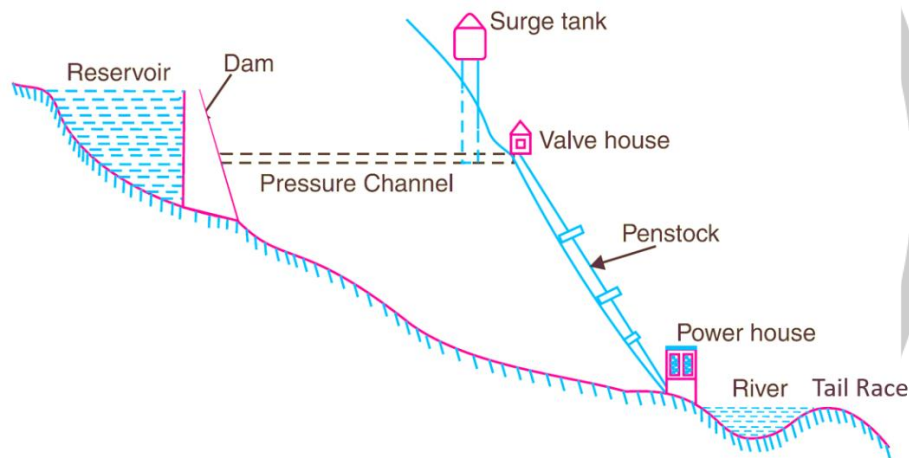


Figure 1.0.4 Schematic diagram of Hydro-electric power plant.

Working and construction of Hydro-electric power plant:

It consists of the following major components:

1. Catchment Area
2. Water Reservoir
3. Dam
4. Valve house
5. Penstock



Catchment Area: It is the area over which rainfall is collected and led to the reservoir.

Water Reservoir: A dam is constructed across a river at a suitable place and water is stored in the catchment area from where it is led to the water reservoir. The main sources of water are rainfall and melting of ice in the mountains.

Dam: It is generally a masonry structure and its functions are to create a water head and to store water in the catchment area. The dam should be able to resist failure against sliding, overturning and rupturing. The water from the reservoir is brought to the surge tank through the pressure channel.

Valve House: At the start of the penstock there is a valve house which contains main sluice Valves for controlling the water flow. In addition to this there are also provided automatic isolating valves which come into operation when the penstock bursts, which cuts off further supply of water.

Penstock: The huge steel or concrete pipe that conveys the water from valve house to water turbine.

Water Turbine: From the penstock water is taken to the turbine. The turbine converts the water energy into mechanical energy.

Surge Tank: It is an open tank built just before the valve house. It acts as the storage of water which is sufficient for one day or for a specific period of time. Surge tank is also helpful in reducing the thrust of water on penstock in case of sudden closing fixed gates of a water turbine.

Pressure Channel: A pressure channel is constructed between reservoir and valve house to carry water from reservoir to valve house.

Alternator: To the turbine is coupled the alternator which converts the mechanical energy into electrical energy.

Tail Race: After the water has done its useful work in the turbine, it is discharged to the tail race which may lead it to the same stream.

Spillway: It is normally constructed at the top of the dam acting as a safety valve by discharging the overflow of water to the down-stream when the reservoir is full during the rainy season. This is generally constructed of concrete and provided with water discharge openings shut off by metal control gates. By changing the degree to which the gates are opened the discharge of water head can be accomplished in order to maintain the water level in the reservoir.

Classification of Hydro-electric power plant:

1. Low head power plant below 60 m.
2. Medium head power above 60 m and below 300 m.

Types of turbines used in Hydro-electric power plant:

1. Impulse turbine,
2. Reaction turbine,
3. Propeller

Advantages of Hydro-electric power plant:

1. It is quite neat and clean.
2. It is economical in large sizes.
3. Running costs are less since no fuel is required.
4. It can be started in no time.
5. It is simple in construction, robust and requires low maintenance.
6. It has higher efficiency and long life.
7. It helps in irrigation and controlling floods.

Disadvantages of Hydro-electric power plant:

1. High capital cost.
2. High cost of transmission lines.
3. Power supply may be affected during the dry season.
4. Skilled and experienced hands are required to build the plant.

Nuclear power generation:

The power plant that is used to warm the water to generate steam, then this steam can be used for rotating huge turbines for generating electricity. These plants use the heat to warm the water which is generated by nuclear fission. So the atoms in the nuclear fission will split into different smaller atoms for generating energy. The nuclear power plant diagram is shown below.

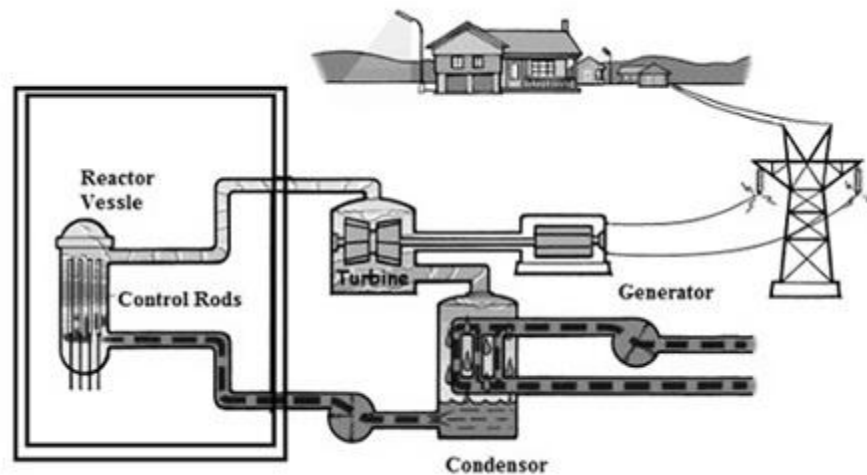


Figure 1.0.5 Nuclear-Power-Plant-Block-Diagram.

Nuclear Power Plant Working Principle

In the power plant, the fission takes place in the reactor and the middle of the reactor is known as the core that includes uranium fuel, and this can be formed into pellets of ceramic. Every pellet generates 150 gallons of oil energy. The total energy generated from the pellets is stacked in metal fuel rods. A bunch of these rods is known as a fuel assembly and a reactor core includes several fuel assemblies.

During nuclear fission, the heat can be generated within the core of the reactor. This heat can be used to warm the water into steam so that turbine blades can be activated. Once the turbine blades activated then they drive the generators to make electricity. In a power plant, a cooling tower is available to cool the steam into the water otherwise they use the water from different resources. Finally, the cooled water can be reused to generate steam.

Components of Nuclear Power Plant

In the above nuclear power plant block diagram, there are different components which include the following.

Nuclear Reactor

In a power plant, a nuclear reactor is an essential component like a heat source that includes the fuel & its reaction of nuclear chain including the waste products of nuclear. The nuclear fuel used in the nuclear reactor is Uranium & its reactions are heat generated in a reactor. Then, this heat can be transferred to the coolant of the reactor to generate heat to all the parts in the power plant.

There are different types of nuclear reactors that are used in the manufacturing of plutonium, ships, satellites & aircraft for research as well as medical purposes. The power plant includes not only includes the reactor and also includes turbines, generators, cooling towers, a variety of safety systems.

Steam Generation

In all the power plants, the production of steam is general; however, the way of generating will change. Most of the plants use water reactors by using two loops of rotating water to generate steam. The primary loop carries very hot water for heating an exchange once water at a low-pressure is circulated, then it warms the water to generate the steam to transmit to the turbine section.

Generator & Turbine

Once the steam is generated, then it travels with high pressures to speed up the turbine. The rotating of the turbines can be used to rotate an electric generator for generating electricity that is transmitted to the electrical grid.

Cooling Towers

In a nuclear power plant, the most essential part is a cooling tower which is used to reduce the heat of the water. Please refer to this link to know more about what is a cooling tower – components, construction & applications

Working of Nuclear Power Plant

The elements like Uranium or Thorium are used in nuclear fission reaction of a nuclear reactor. Because of this fission, a huge amount of heat energy can be generated and it is transmitted to the coolant reactor. Here, the coolant is nothing but water, liquid metal otherwise gas. The water is heated to flow in a heat exchanger so that it changes into high-temperature steam. Then the steam which is produced is permitted to make a steam turbine run. Again the steam can be changed back into the coolant & recycled to use for the heat exchanger. So, the turbine and alternator are connected to produce electricity. By using a transformer, the electricity which is produced can be increased to use in long-distance communication.

Efficiency of the Nuclear Power Plant

The nuclear power plant efficiency can be decided equally to other heat engines because technically the plant is a large heat engine. The sum of electric power generated for every unit of thermal power will provide the plant is thermal efficiency & because of the thermodynamics second law, there is a higher limit to how efficient these power plants can be.

The normal nuclear power plants attain efficiencies approximately 33 to 37%, equivalent to fossil-fueled plants. High temperature & more current designs such as the Generation IV reactors could acquire above 45% efficiency.

Types of Nuclear Power Plant

There are two types of nuclear power plants such as pressurized water reactor and boiling water reactor.

Pressurized Water Reactor

In this kind of reactor, regular water is used as a coolant. This is kept at extremely high force so that it does not get a boil. A heat exchanger in this reactor transfers the heated water where the water from the secondary coolant circle is changed into vapor. Therefore, this loop is totally

free from the material of radioactive. In this reactor, the coolant water works as a moderator. Because of these benefits, these reactors are used most frequently.

Boiling Water Reactor

In this kind of reactor, a single coolant loop is only available. The water is permissible to heat within the reactor. The steam is produced from the reactor when it heads out from the reactor & the steam will flow throughout the steam turbine. The main drawback of this reactor is, the coolant water approaches the fuel rods & the turbine. So, radioactive material could be located over the turbine.

Site Selection for Nuclear Power Plant

The selection of the site for nuclear PowerPlant can be done by considering the technical requirement. The arrangement and working of a nuclear power plant mainly depend on the characteristics of the site.

While designing the plant, the risks from the site must be considered. The plant design has to handle with tremendous natural occurrence & human-induced actions, without damaging the operational security of the plant.

Each site has to give needed necessities like discarded and decay heat sinks, power supply availability, excellent communications and efficient crisis management, etc. For a power plant, the estimate of the site typically occupies different stages like selection, characterization, pre-operational, and operational.

Nuclear Power Plants in India

There seven nuclear power plants in India which include the following.

Kudankulam Nuclear Power Plant, located in Tamil Nadu

Tarapur Nuclear Reactor, located in Maharashtra

Rajasthan Atomic Power Plant, located in Rajasthan

Kaiga Atomic Power Plant, located in Karnataka

Kalapakkam Nuclear Power Plant, located in Tamil Nadu

Narora Nuclear Reactor, located in Uttar Pradesh

Kakarapar Atomic Power Plant, located in Gujarat

Advantages

The advantages of nuclear power plants include the following.

- It uses less space compared with other power plants
- It is extremely economical and generates huge electric power.
- These plants are located near the load center because there is no requirement of huge fuel.
- It generates a huge amount of power in the process of each nuclear fission
- It uses less fuel to generate huge energy
- Its operation is reliable
- When compared with steam power plants, it is very clean and neat
- The operating cost is small
- It doesn't produce polluting gases

Disadvantages

The disadvantages of nuclear power plants include the following.

- The cost of primary installation is extremely high when compared with other power stations.
- The nuclear fuel is expensive so recovering is difficult
- High capital cost compare with other power plants
- Technical knowledge is required to operate this plat. So maintenance, as well as salary, will be high.
- There is a chance of radioactive pollution
- The response is not efficient
- The requirement of cooling water is double compare with a steam power plant.

Applications

The applications of nuclear power plants include the following.

Nuclear energy is used in different industries all over the world for desalination of ocean water, production of hydrogen, district cooling/heating, the removal of tertiary oil resources & used in heat process applications like cogeneration, conversion of coal to liquids & help in the chemical feedstock synthesis.

Wind power generation

Wind energy is a natural form of energy that is capable of producing electrical or mechanical forces. Windmills or wind turbines are devices that are capable of converting the kinetic energy of wind into mechanical energy. This mechanical energy is further converted into electrical energy.

Importance of Wind Energy

The following are the important features of Wind Energy:

- Wind energy is environment-friendly.
- The cheapest source of electrical energy.
- A project of wind energy is the fastest payback period.
- Operation and maintenance costs are low.
- A wind energy project is no investment in manpower.
- A wind energy project is a fast-track power project with a lower gestation (reproductive cycle) period and a modular concept.
- The cost per kWh reduces over a period of time as against rising conventional power projects.
- Wind energy is plentiful throughout the world.
- During the production of this energy, no pollution of air or water occurs.
- Necessary machines for harnessing this power are available economically.
- We are going to discuss the most essential parts of wind turbine.

Parts of Wind Turbine

Following are the different parts of the wind turbine:

1. Blades
2. The rotor
3. Nacelle
4. A gearbox and coupling (transmission system)
5. Aero turbine
6. Controller

7. Electrical generator
8. Supporting structure.

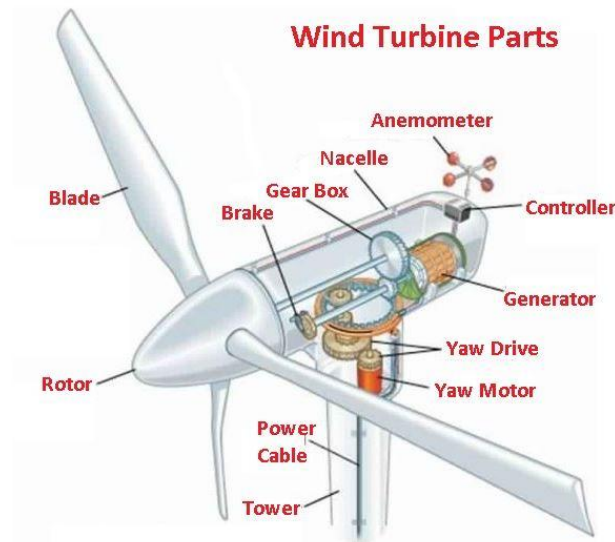


Figure 1.6 Wind turbine parts.

1 Blade

Lifting-style wind turbine blades. These are designed most efficiently, especially to capture the energy of strong, fast winds. Some European companies actually manufacture single-blade turbines.

2 The Rotor

The rotor is aerodynamically designed to occupy the maximum surface area of the wind to spin the most ergonomically. The blades are lightweight, durable, and corrosion-resistant material. The best materials are composites of fiberglass and reinforced plastic.

3 Nacelle

A housing that contains all the components which are essential to operate the turbine efficiently is called a nacelle. It is mounted on top of a tower and includes gearboxes, generators, low and high-speed shafts, controllers, and brakes.

The nacelle is fitted with a wind speed anemometer and a wind vane. Nacelle provides housing for:

Unit 1

1. Low-speed shaft
2. Brake
3. Gearbox
4. High-speed shaft
5. Generator
6. Anemometer
7. Wind vane.



4 A Gearbox and Coupling (Transmission System)

A gearbox magnifies or amplifies the energy output of the rotor. The gearbox is located directly between the rotor and the generator. A rotor rotates the generator as directed by the tail vane.

5 Aeroturbine

Aeroturbine converts wind energy into rotary mechanical energy.

6 Controller

The controller realizes the wind direction wind speed output of the generator rotor and other required performance quantities of the system and initiates appropriate control signals to take appropriate corrective action.

7 Electrical Generator

This unit produces electricity from the rotation of the rotor. The generator comes in various sizes with respect to the output. This generator converts mechanical energy into electrical power. The output of the generator is coupled to the load or system grid.

8 Supporting Structure

This is the heavy structure set up with a proper foundation and carries all the components of the windmill. It should be properly designed with a proper factor of safety to withstand a dead load of all components and wind force.

Working of Wind Power Plant

The wind turbines or wind generators use the power of the wind which they turn into electricity. The speed of the wind turns the blades of a rotor (between 10 and 25 turns per minute), a source of mechanical energy. The rotor then turns on a generator that converts mechanical energy into electricity.

Unit 1

As the wind blows, a wind turbine converts the kinetic energy of the wind's motion into mechanical energy by the rotation of the rotor, and this mechanical energy is transmitted by the shaft to the generator through the gear train. The generator converts this mechanical energy into electrical energy, thereby generating electricity.

An electric motor orientates the nacelle so that its rotor is placed facing the wind. Each turbine is made up of a mast between 20 m and 100 m according to the power of the machines. The mast supports the rotor usually equipped with three blades, and the nacelle which contains the generator with electrical and mechanical backup.

Wind turbines are designed for wind speeds of between 14 and 90 km/hour. Above that, the braking mechanism automatically stops the turbine for the safety of the equipment and to minimize wear and tear. Modern wind turbines supply their normal power at around 50 km/h.

A wind turbine is connected to the electricity network via a transformer located at the base of the mast. The electricity produced is generally stepped up to the voltage of the network (20 kV), then it passes through a delivery substation before being fed into the distribution or electrical transport network.

Factors Affecting the Distribution of Wind Energy

The power output of the wind turbines can be increased by turning the head in such a way that the blades face the wind, this can be done with a wind direction sensor connected with a motorized head moving mechanism (or yaw).

The following are some factors that affect the distribution of wind energy:

- Wind speed.
- Good road access to the site.
- Suitable region and geology for onsite access.
- Low pollution density.
- Minimum risk of agroforestry operations.
- Nearness to the electrical grid.
- Supportive landholders.
- Privately owned freehold land.
- Good industrial support for construction and ongoing operation.
- High annual average wind speed.
- Height of the windmill.

Unit 1

- The altitude of the site.
- A smooth steady wind speed that should blow all the time.
- Government policy.
- Demand for electricity.
- Icing problems, salt spray, or blowing dust should be present at the site.
- Blade aerodynamics.
- Obstructions
- Air temperature.
- Performance of wind turbine.



Applications of Wind Power

The following are the applications of wind energy:

- It is used for pumping water.
- It is used for generating electricity.
- Used for running the floor mill.
- It is used to lift the water for irrigation purposes.
- It is used for milling and grinding applications.

Advantages of Wind Energy or Wind Power Plant

The following are the advantages of wind power plants:

- Wind energy is a renewable energy source.
- It does not require any fuel and avoids transportation.
- Being free from pollution helps in maintaining ecological balance.
- It is very economical and competitive.
- High-altitude areas which are isolated from public use can be utilized for setting the windmill.
- A small plot of land is required.
- It does not require water.
- Free from risks associated with volatile fossil fuel.

Disadvantages of Wind Power Plant

The following are the disadvantages of wind power plant:

- Continuous power generation is not possible due to fluctuation
- Noisy is in operation during the conversion of energy.

Unit 1

- It requires a larger area.
- It needs high storage capacity due to its irregularity.
- Wind energy systems have a high overall weight
- These systems require proper maintenance.

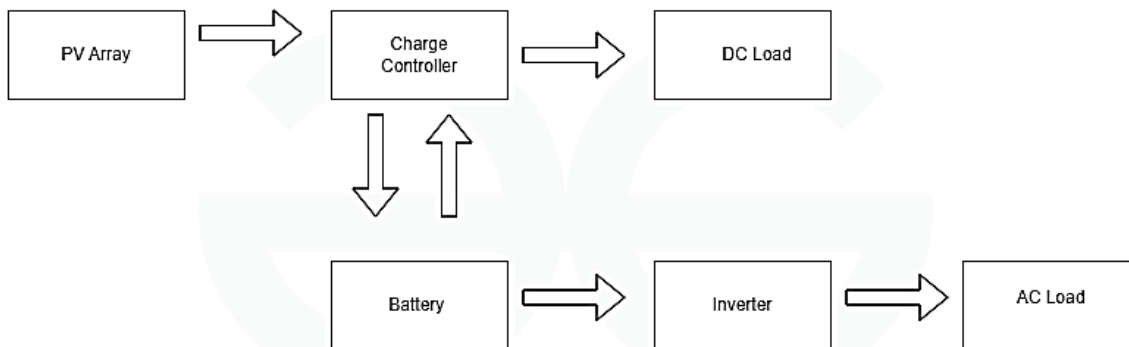
Solar power generation

Solar energy is a renewable and sustainable form of power derived from the radiant energy of the sun. This energy is harnessed through various technologies, primarily through photovoltaic cells and solar thermal systems. Photovoltaic cells commonly known as solar panels, convert sunlight directly into electricity by utilizing the photoelectric effect. These cells are typically made of semiconductor materials, such as silicon, which release electrons when exposed to sunlight.

Solar thermal systems, on the other hand, capture the sun's heat to produce steam, driving turbines that produce electricity. This technology often involves mirrors or lenses to concentrate sunlight onto a small area, intensifying the heat.

Block Diagram of Solar Energy

A solar energy block diagram illustrates the key components and their interconnections in solar power systems. Here's a simplified explanation of the main components typically found in such a diagram:



Solar panels (photovoltaic modules): Solar panels are the primary components that capture sunlight and convert it into electrical energy through the photovoltaic effect. These panels are made up of semiconductor materials like silicon.

Inverter: The direct current electricity generated by solar panels needs to be converted into alternating current to be compatible with most electrical appliances and the grid. The inverter performs this conversion.

Charge controller: In off grid solar systems with energy storage, a charge controller is used to regulate the charging and discharging of the batteries. It prevents overcharging and excessive discharging, extending the life span of the batteries.

Battery bank: In off-grid or hybrid solar systems energy storage is often required to store excess electricity generated during sunny periods for use during cloudy days or at night. A battery bank stores this energy for later use.

Load: The load represents the electrical appliances or devices that consume the electricity generated by the solar power system. This can include household appliances, lighting and other electrical equipment.

Uses of Solar Energy

Solar furnaces: Solar furnaces must operate at extremely high temperatures. In this method, solar radiation requires slanted, rotating mirrors to generate high heat.

Solar green-houses: Solar greenhouses keeps harmful external elements away from plants, bringing positive CO₂ air inside instead. Glass or plastic also covers the greenhouse to maintain controlled indoor temperatures and provide an ideal growing environment for plants year-round.

Advantages and Disadvantages of Solar Energy

There are some lists of Advantages and Disadvantages of Solar Energy given below:

Advantages of Solar Energy

Solar energy is a clean and renewable energy source harnessing power from the sun without producing harmful pollutants or greenhouse gases.

Solar power allows individuals, business and communities to generate their own electricity, leading to reduced dependence on traditional utility grids.

Solar energy reduces the dependence on fossil fuels and foreign energy sources.

Once installed, solar panels have relatively low operating and maintenance costs.

Unit 1

Distributed solar power generation can enhance grid stability by reducing the need for centralized power plants and long-distance transmission lines.

Solar energy systems are scalable and adaptable to various needs from small installations to large utility scale solar forms.

Ongoing research and development in solar technology continue to drive advancements, improving efficiency and reducing costs. Innovations such as thin-film solar cells, solar tracking systems and energy storage solutions enhance the overall effectiveness of solar energy systems.

Disadvantages of Solar Energy

Solar energy production is dependent on sunlight, making it intermittent. Energy generation is significantly reduced during cloudy days, night time and in regions with frequent inclement weather.

The installation of solar panels and associated equipment can involve high upfront costs. While the prices are reducing over the years but still can be a barrier for individuals or business.

Storing excess energy for use during periods of low sunlight can be challenging.

Applications of Solar Energy

Solar electricity: Photovoltaic cells generate electricity through direct sunlight. There are various electrical benefits to using solar electric power generation such as reliability, low maintenance costs, durability and eco-friendly. It is more beneficial for irrigation, commercial grid power systems and more.

Solar distillation: Solar distillation utilizes abundant sunlight to convert saline water into distilled water. Solar radiation turns into heat, producing purified water for cooling. This method offers cost effective production of distilled water, unlike traditional electrical avenues.

Solar water heating: Solar water heating is an eco-friendly alternative to traditional heaters, employs active systems like direct and indirect circulation. Passive water systems in solar water heating involving integral collector storage and thermosiphon systems. These methods harness solar energy efficiently, promoting sustainability in water heating.

Ohm's Law and Power

Ohm's Law

It can be determined, for many conductors, that current is proportional to potential difference across the conductor.

$$I \propto V$$

Also, the current in a wire is inversely proportional to the wire's resistance.

$$I \propto \frac{1}{R}$$

George Simon Ohm (1787 – 1854) noted that the current (I) in a conductor is directly proportional to the applied voltage (V). The proportionality constant was later determined to be the resistance (R), which gives the following equation known as Ohm's law.

Ohm's Law

$$V = IR$$

Equation 9

It is important to note that Ohm's law is based on experimental observation, and it is not a fundamental law of nature. Therefore, not all materials will obey Ohm's law. Materials that obey Ohm's law are called ohmic and those that do not are called non-ohmic.

Example 4

An object carries a current of 6 A when connected to a 120 V source. What is the resistance of the object?



Solution:

Using Ohm's law from Equation 9 gives

$$V = IR$$

$$R = \frac{V}{I} = \frac{120 \text{ V}}{6 \text{ A}}$$

$$R = 20 \Omega$$

Power

The amount of work that can be accomplished in a specified amount of time is called power. Work has units of joules (J), and time has units of seconds (s). Therefore, power has units of joules per second. For electrical work, the unit is defined as a watt (W). One watt is equal to 1 joule of work that is done in 1 second. Without derivation, equations for electrical power are given below.

$$P = IV$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

Power

Equation 10

Example 5

What power is delivered to a DC motor when the input current is 10 A and the supply voltage is 120 V?



Solution:

From Equation 10

$$P = IV = 10 (120 V)$$

$$P = 1,200 W$$

Example 6

How much does it cost to burn a 60 W lightbulb for 8 hours if electricity costs 12 cents per kilowatt-hour?

Solution:

The energy consumed will equal the power times time.

$$E = (0.06 kW)(8 hr) = 0.48 kWh$$

At 12 cents per kilowatt-hour, the cost is

$$Cost = 0.48 kWh(\$0.12/kWh)$$

$$Cost = \$0.06$$

Single and Multi-Loop Circuits

Introduction

This section will provide foundational information dealing with circuit analysis. Content is divided into two categories: single-loop circuits, and multi-loop circuits. Single-loop circuits have all resistors arranged in series, while multi-loop circuits contain all resistors arranged in parallel. Circuits containing a combination of series and parallel arrangements will be discussed in Section 6.4.

Single-Loop Circuits (Series Circuits)

Resistors in Series

Figure 4 (a) shows a series circuit with two resistors. For a series circuit, there is only one pathway for the current.

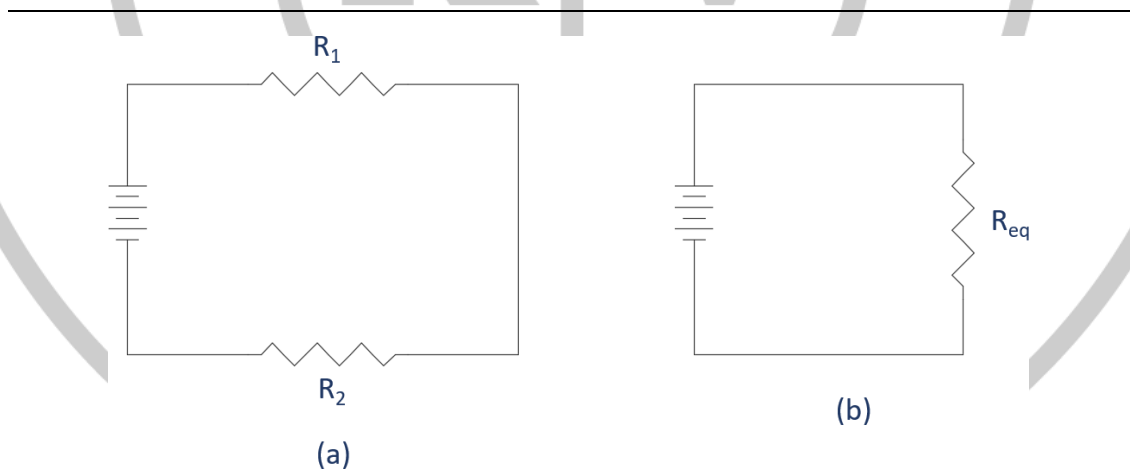


Figure 4 (a) Two resistors in series (b) Circuit with equivalent resistance

Therefore, current is the same through each resistor in a series circuit. The potential drop through the first resistor is equal to IR_1 , and the potential drop through the second resistor is equal to IR_2 . Therefore, the potential drop across both will be

$$V = IR_1 + IR_2$$

Equation 11

Figure 4 (b) shows the circuit with a single equivalent resistor with a resistance R_{eq} . Applying Ohm's law to the equivalent resistor gives

$$V = IR_{eq}$$

Equation 12

Equating Equation 11 and Equation 12 gives

$$IR_{eq} = IR_1 + IR_2$$

$$R_{eq} = R_1 + R_2$$

Equation 13

Therefore, resistors in series can be replaced by an equivalent resistor with a resistance equal to the summation of all resistors. For N resistors in series the equivalent resistance is calculated using

$$R_{eq} = R_1 + R_2 + \dots + R_N$$

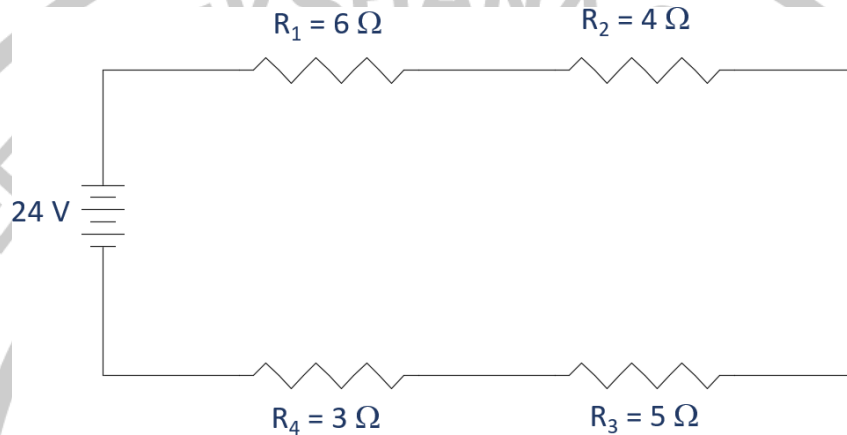
N

$$R_{eq} = \sum_{i=1}^N R_i$$

Equation 14

Example 7

A circuit contains 4 resistors in series. What is the current in the circuit?



Solution:

The equivalent resistance is calculated using Equation 14.

$$R_{eq} = \sum_{i=1}^N R_i = 6\ \Omega + 4\ \Omega + 5\ \Omega + 3\ \Omega = 18\ \Omega$$

Ohm's law, from Equation 12, can now be used to calculate current.

$$V = IR_{eq}$$
$$I = \frac{V}{R_{eq}} = \frac{24\ V}{18\ \Omega}$$

$$I = 1\frac{1}{3}\ A$$

Kirchoff's Voltage Law

In single-loop circuits (series circuits), the algebraic sum of the changes in potential encountered in the loop must equal zero. This law, which is referred to as Kirchoff's loop rule or Kirchoff's voltage law (named after German physicist Gustav Robert Kirchoff), is illustrated graphically in Figure 5.

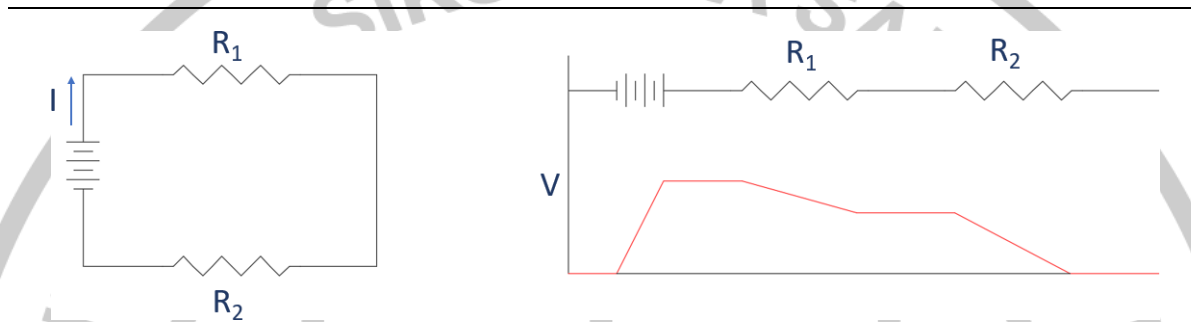


Figure 5 Voltage change through the circuit

Another way to word the voltage law is that the sum of all voltage drops in a circuit must equal the applied voltage. For a series of N resistors in a circuit, this can be expressed mathematically as

$$V_{supply} = V_{R1} + V_{R2} + V_{R3} + \dots + V_{RN} \quad \text{Equation 15}$$

Multi-Loop Circuits (Parallel Circuits)

Resistors in Parallel

Consider two resistors connected in parallel, as shown in Figure 6 (a). The potential difference must be the same across each resistor when arranged in parallel. The current, in general, will not be the same in each resistor (it will only be the same if the resistors have equal resistance).

Introduction to Electrical Theory and DC Circuits

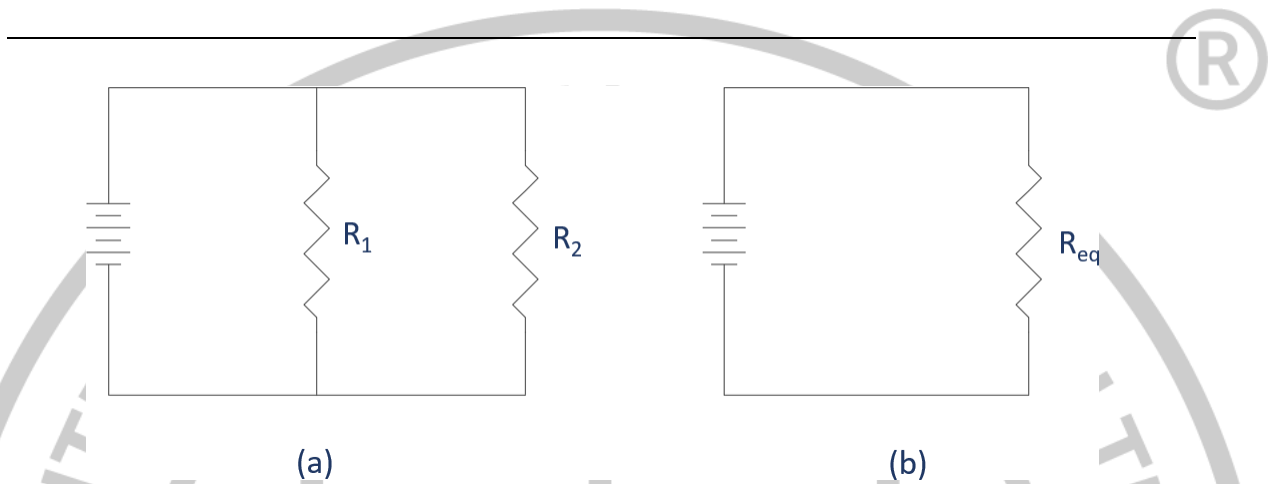


Figure 6 (a) Two resistors in parallel (b) Circuit with equivalent resistance

1.1.1 Kirchoff's Current Law

Kirchoff's current law states that the sum of the currents entering any junction must be equal to the sum of the currents leaving that junction. The concept is illustrated in Figure 7.

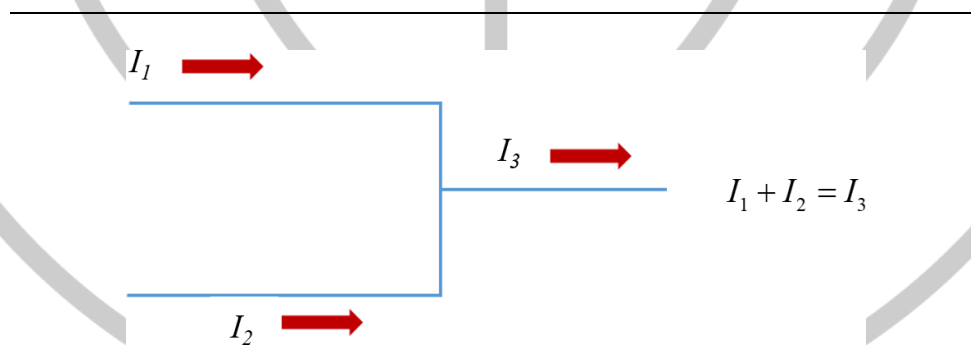


Figure 7 Kirchoff's current law

The resistors in parallel can be replaced by an equivalent resistor as shown in Figure 6 (b). Applying Ohm's law to the equivalent resistor give

$$I = \frac{V}{R_{eq}} \quad \text{Equation 16}$$

Equation 16 can be substituted into Kirchhoff's current law, which is shown in Figure 7, to give

$$\begin{aligned} \frac{V}{R_{eq}} &= \frac{V}{R_1} + \frac{V}{R_2} \\ \frac{1}{R_{eq}} &= \frac{1}{R_1} + \frac{1}{R_2} \end{aligned} \quad \text{Equation 17}$$

Therefore, the equivalent resistance for N resistors arranged in parallel equals

$$\frac{1}{R_{eq}} = \sum_{i=1}^N \frac{1}{R_i} \quad \text{Equation 18}$$

From Equation 18 it can be determined that the equivalent resistance of two or more resistors connected in parallel will always be less than the smallest resistor in the group.

Three resistors, which resistances of $3\ \Omega$, $6\ \Omega$, and $9\ \Omega$, are connected in parallel. What is the equivalent resistance?

Solution:

From Equation 18

$$\frac{1}{R_{eq}} = \sum_{i=1}^N \frac{1}{R_i} = \frac{1}{3\ \Omega} + \frac{1}{6\ \Omega} + \frac{1}{9\ \Omega}$$

$$R_{eq} = 1.64\ \Omega$$

Further examples of parallel circuits will be provided in Section 6.3.

DC Circuit Analysis

Introduction

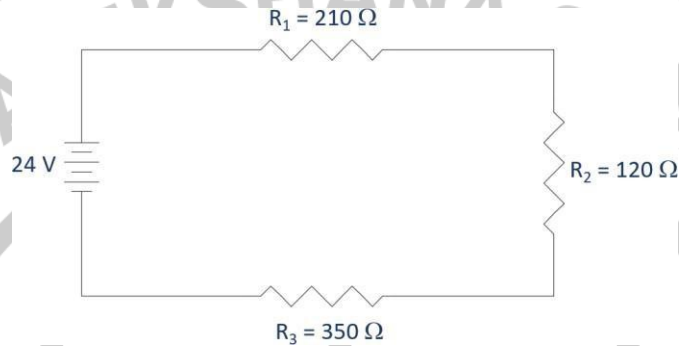
This section will provide application examples of series and parallel DC circuits. The concepts covered in previous sections will be used to analyze the circuits. In addition, some new concepts will be introduced to aid in the analysis of the circuits. Application examples will be separated into categories of series circuits and parallel circuits. The last section will consider problems that are a combination of series and parallel arrangements.

Series Circuits

We will begin with examples involving only series circuits.

Example 9

Determine the amount of current that will flow for the circuit shown. What is voltage drop across resistor R_2 ?



Solution:

The total equivalent resistance can be calculated using Equation 14.

$$R_{eq} = R_1 + R_2 + R_3$$

$$R_{eq} = 210 \Omega + 120 \Omega + 350 \Omega = 680 \Omega$$

Rearranging Equation 12 will give the current.

$$V = IR_{eq}$$

$$I = \frac{V}{R_{eq}} = \frac{24 V}{680 \Omega}$$

$$I = 35.29 \text{ mA}$$

The voltage drop across resistor R_2 can be determined by multiplying the current and the resistance.

$$V_{R2} = IR_2 = 0.03529 (120 \Omega)$$

$$V_{R2} = 4.24 V$$

Voltage Divider Rule

The second part of the previous example asked for the voltage drop across a resistor. The method used required the current. However, another method exists for calculating voltage drops across a given resistor in a series circuit. The **voltage divider rule** states that the ratio between any two voltage drops in a series circuit is equal to the ratio of the two resistances.

Mathematically, the voltage divider rule states

$$V_{Rx} = V_T \frac{R_x}{R_T} \quad \text{Equation 19}$$

Example 10

Use the voltage divider rule to determine the voltage drop across each resistor in the previous example.

Solution:

Using Equation 19 for each resistor gives

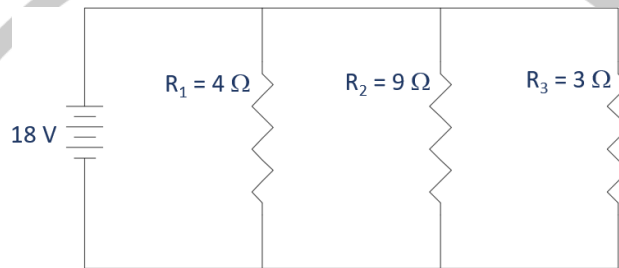
$$\begin{aligned}
 V_{R1} &= V_T \frac{R_1}{R_T} = 24 \text{ V} \frac{210 \Omega}{680 \Omega} = 7.41 \text{ V} \\
 V_{R2} &= V_T \frac{R_2}{R_T} = 24 \text{ V} \frac{120 \Omega}{680 \Omega} = 4.24 \text{ V} \\
 V_{R3} &= V_T \frac{R_3}{R_T} = 24 \text{ V} \frac{350 \Omega}{680 \Omega} = 12.35 \text{ V}
 \end{aligned}$$

Parallel Circuits

Now we can consider examples only using parallel arrangements.

Example 11

Three resistors are connected in parallel as shown. Determine the current in each resistor.



Solution:

For a parallel circuit, the voltage will be the same through each resistor. Using Ohm's law gives

$$I_1 = \frac{V}{R_1} = \frac{18\text{ V}}{4\ \Omega} = 4.5\text{ A}$$

$$I_2 = \frac{V}{R_2} = \frac{18\text{ V}}{9\ \Omega} = 2\text{ A}$$

$$I_3 = \frac{V}{R_3} = \frac{18\text{ V}}{3\ \Omega} = 6\text{ A}$$

Unit 1

The concept of parallel circuits can be treated the same as fluid flow in a branch. Fluid will follow the path of least resistance. Notice that the flow (current) followed the path of least resistance because the largest current flow was through the lowest resistor.

Next, we will consider a problem using Kirchoff's current law discussed in Section 5.3.2. The current law was briefly introduced in Section 5.3.2, but it will be expanded here. It is important to understand the sign convention. The four rules below apply, with each assuming we are moving from point *a* to point *b*.

- 1 If a resistor is traversed in the direction of the current:

$$\Delta V = V_b - V_a = -IR$$

- 2 If a resistor is traversed in the direction opposite the current:

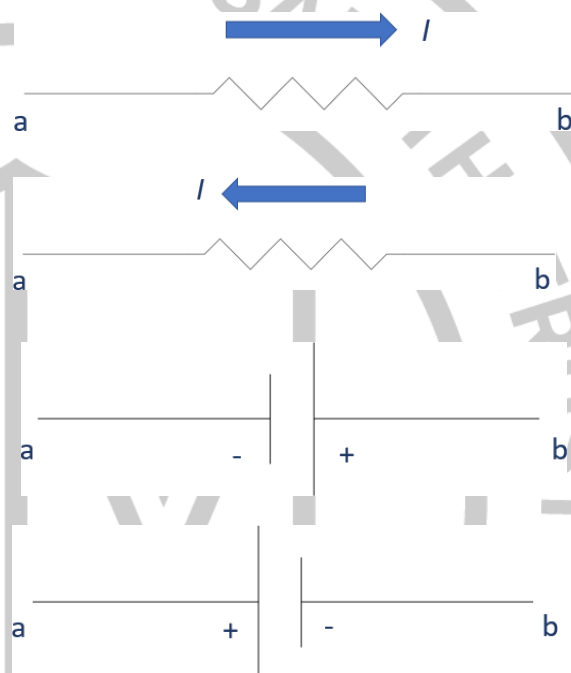
$$\Delta V = V_b - V_a = +IR$$

- 3 If a voltage source is traversed from - to +:

$$\Delta V = V_b - V_a = +V$$

- 4 If a voltage source is traversed from + to -:

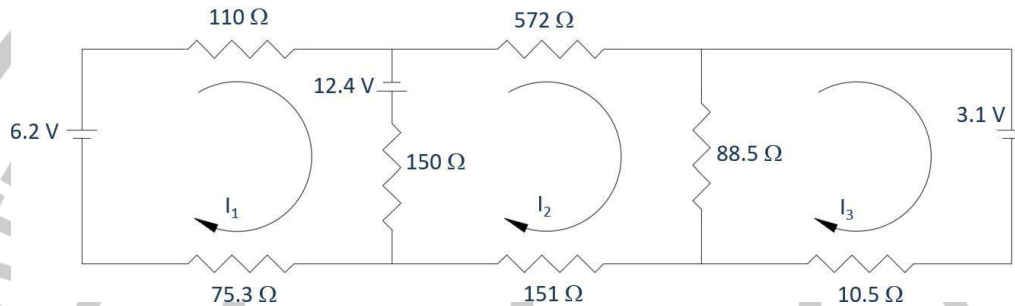
$$\Delta V = V_b - V_a = -V$$



The rules, and the process of using Kirchoff's current law, are best illustrated in an example problem. For each loop in the following example use the rules above to determine the proper sign.

Example 12

Use Kirchoff's law to determine the value of the three currents.



Solution:

Loop 1

$$6.2 - 110I_1 + 12.4 - 150I_1 + 150I_2 - 75.3I_1 = 0$$

$$-335I_1 + 150I_2 = -18.6 \quad [1]$$

Loop 2

$$-150I_2 + 150I_1 - 12.4 - 572I_2 - 88.5I_2 + 88.5I_3 - 151I_2 = 0$$

$$150I_1 - 962I_2 + 88.5I_3 = 12.4 \quad [2]$$

Loop 3

$$-88.5I_3 + 88.5I_2 - 3.1 - 10.5I_3 = 0$$

$$88.5I_2 - 99I_3 = 3.1 \quad [3]$$

Putting equations [1], [2], and [3] in matrix form gives

$$\begin{bmatrix} -335 & 150 & 0 \\ 150 & -962 & 88.5 \\ 0 & 88.5 & -99 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -18.9 \\ 12.4 \\ 3.1 \end{bmatrix}$$

Solving the system of equations for the current values gives

$$I_1 = 52.7 \text{ mA}$$

$$I_2 = -8.22 \text{ mA}$$

$$I_3 = -38.7 \text{ mA}$$



Current Divider Rule

The current divider rule states that amount of current in one of two parallel resistances is equal to the product of total current and other resistance divided by the sum of resistance.

Current Divider Rule

$$I_1 = \frac{I_T R_2}{R_1 + R_2}$$

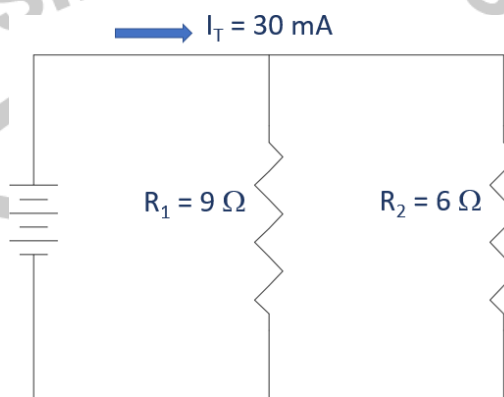
$$I_2 = \frac{I_T R_1}{R_1 + R_2}$$

Equation 20

Example 13



Use the current divider rule to determine the currents flowing through R_1 and R_2 .



Solution:

Using the current divider rule in Equation 20

$$I_1 = \frac{I_T R_2}{R_1 + R_2} = \frac{30 (6 \Omega)}{(9 \Omega) + (6 \Omega)}$$

$$I_1 = 12 \text{ mA}$$

$$I_2 = \frac{I_T R_1}{R_1 + R_2} = \frac{30 (9 \Omega)}{(9 \Omega) + (6 \Omega)}$$

$$I_2 = 18 \text{ mA}$$

Series-Parallel Circuits

Circuits that contain a combination of series and parallel circuits are known as series-parallel circuits. The equations for equivalent resistance for series and parallel circuits can be applied to systems with a combination of series and parallel arrangements. Consider the arrangement shown in Figure 8. The $5\ \Omega$ and $9\ \Omega$ resistors are arranged in series, and they can be simplified to the $14\ \Omega$ equivalent resistor using Equation 14. The $6\ \Omega$ and $3\ \Omega$ resistors are arranged in parallel, and they can be simplified to the equivalent resistance of $2\ \Omega$ using Equation 18. The new arrangement has two resistors ($14\ \Omega$ and $2\ \Omega$) in series, which can be simplified again to the final equivalent resistance of $16\ \Omega$.

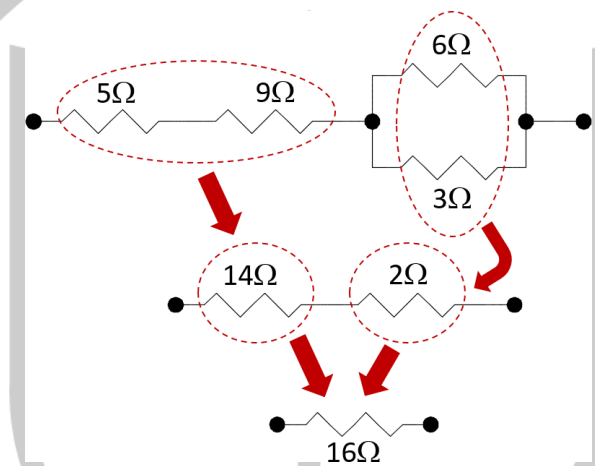
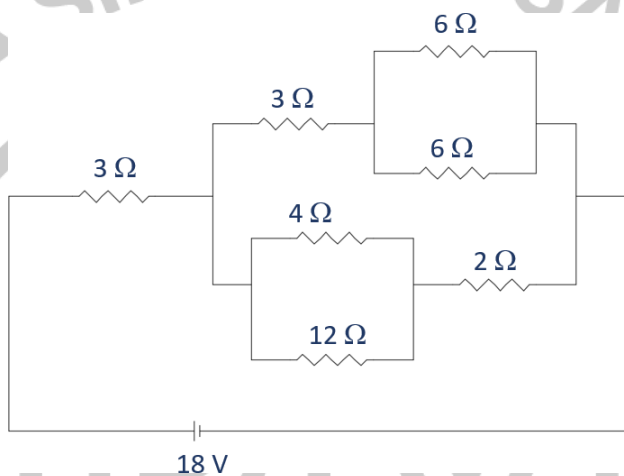


Figure 8 Equivalent resistance concept for series-parallel circuits

Example 14

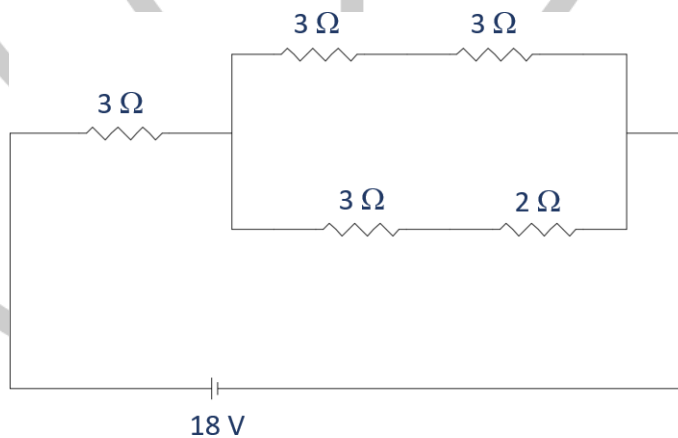


Determine the current in the $12\ \Omega$ resistor for the circuit shown.

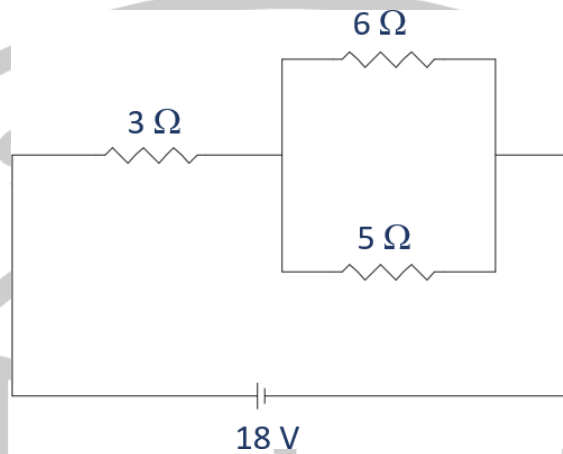


Solution:

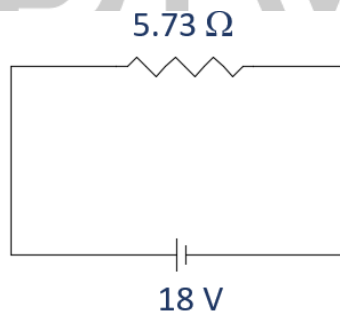
The first step is to determine the equivalent resistance for all resistors. The two $6\ \Omega$ resistors and the $4\ \Omega$ and $12\ \Omega$ resistors in parallel can be reduced using Equation 18.



The resulting parallel circuits now have resistors in series, which can be simplified using Equation 14.



The two resistors in parallel can be reduced to one 2.73 Ω, which leaves two resistors in series. Summing the two in series gives the final equivalent resistance.



Ohm's law can now be used to determine the total current.

$$I_T = \frac{V}{R_{eq}} = \frac{18\text{ V}}{5.73\text{ }\Omega} = 3.14\text{ A}$$

Move now to the previous figure. The full 3.14 A will pass through the 3 Ω resistor and will be split between the two in parallel. The current divider rule can be used to determine the current passing through the 5 Ω resistor.

$$I_{5\Omega} = \frac{I_T R_1}{R_1 + R_2} = \frac{3.14 (6\Omega)}{6\Omega + 5\Omega} = 1.71 \text{ A}$$

Now reference the original figure. The 1.71 A will be split between the 4 Ω and 12 Ω resistors. Using the current divider rule to determine the current through the 12 Ω resistor gives

